

Table 54. ADC characteristics⁽¹⁾ (continued)

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
C _{ADC}	Internal sample and hold capacitor	-	-	5	-	pF
t _{STAB}	ADC power-up time	LDO already started	2			Conversion cycle
t _{CAL}	Calibration time	f _{ADC} = 35 MHz	2.35			μs
		-	82			1/f _{ADC}
t _{LATR}	Trigger conversion latency for regular and injected channels without aborting the conversion	CKMODE = 00	2	-	3	1/f _{ADC}
		CKMODE = 01	6.5			1/f _{PCLK}
		CKMODE = 10	12.5			
		CKMODE = 11	3.5			
t _s	Sampling time	f _{ADC} = 35 MHz	0.043	-	4.59	μs
			1.5	-	160.5	1/f _{ADC}
t _{ADCVREG_STUP}	ADC voltage regulator start-up time	-	-	-	20	μs
t _{CONV}	Total conversion time (including sampling time)	f _{ADC} = 35 MHz Resolution = 12 bits	0.40	-	4.95	μs
		Resolution = 12 bits	t _s + 12.5 cycles for successive approximation = 14 to 173			1/f _{ADC}
t _{IDLE}	Laps of time allowed between two conversions without rearm	-	-	-	100	μs
I _{DDA(ADC)}	ADC consumption from V _{DDA}	f _s = 2.5 MSps	-	410	-	μA
		f _s = 1 MSps	-	164	-	
		f _s = 10 kSps	-	17	-	
I _{DDV(ADC)}	ADC consumption from V _{REF+}	f _s = 2.5 MSps	-	65	-	μA
		f _s = 1 MSps	-	26	-	
		f _s = 10 kSps	-	0.26	-	

1. Specified by design – Not tested in production.

2. V_{REF+} is internally connected to V_{DDA} on some packages. Refer to [Section 4: Pinouts, pin description and alternate functions](#) for further details.